some of which get deposited on the stigma of the flower when it visits next.

c. Malacophily: Pollination by slugs and snails is called malacophily. Some plants of Araceae are pollinated by snails. Water snails crawling among *Lemna* pollinate them.

d. Entomophily: Pollination by insects is called **Entomophily**. Pollination by ants is called **myrmecophily**. Insects that are well adapted to bring about pollination are bees, moths, butterflies, flies, wasps and beetles. Of the insects, bees are the main flower visitors and dominant pollinators. Insects are chief pollinating agents and majority of angiosperms are adapted for insect pollination. It is the most common type of pollination.

The characteristic features of entomophilous flowers are as follows:

- Flowers are generally large or if small they are aggregated in dense inflorescence. Example: Asteraceae flowers.
- Flowers are brightly coloured. The adjacent parts of the flowers may also be brightly coloured to attract insect. For example in *Poinsettia* and *Bougainvillea* the bracts become coloured.
- Flowers are scented and produce nectar.
- Flowers in which there is no secretion of nectar, the pollen is either consumed as food or used in building up of its hive by the honeybees. Pollen and nectar are the floral rewards for the visitors.
- Flowers pollinated by flies and beetles produce foul odour to attract pollinators.
- In some flowers juicy cells are present which are pierced and the contents are sucked by the insects.

Pollination in *Salvia* (Lever mechanism):

The flower is protandrous and the corolla is bilabiate with 2 stamens. A lever mechanism helps in pollination. Each anther has an upper fertile lobe and lower sterile lobe which is separated by a long connective which helps the anthers to swing freely. When a bee visits a flower, it sits on the lower lip which acts as a platform. It enters the flower to suck the nectar by pushing its head into the corolla. During the entry of the bee into the flower the body strikes against the sterile end of the connective. This makes the fertile part of the stamen to descend and strike at the back of the bee. The pollen gets deposited on the back of the bee. When it visits another flower, the pollen gets rubbed against the stigma and completes the act of pollination in *Salvia* (Figure 1.17a). Some of the other interesting pollination mechanisms found in plants are a) Trap mechanism (*Aristolochia*);Pit fall mechanism (*Arum*);Clip or translator mechanism (*Asclepiadaceae*) and Piston mechanism (*Papilionaceae*).

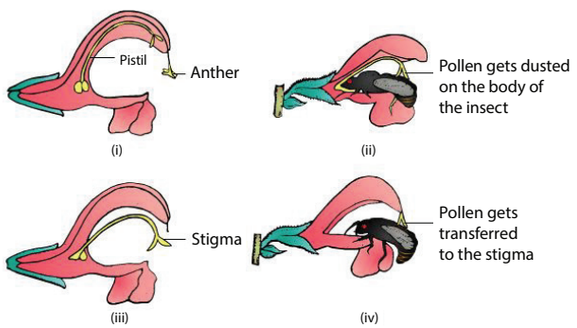


Figure 1.17 (a) Pollination in *Salvia* - Lever mechanism

Pollination in *Calotropis* (Translator mechanism)

This mechanism is found in members of Asclepiadaceae(Apocynaceae as per APG system of classification). The flowers are bisexual with 5 stamens forming **gynostegium**(union of stigma with the androecium). The stigma is large and 5 – angled and is receptive on the underside. Each stamen at its back possesses a brightly coloured hood like outgrowth enclosing horn shaped nectar. The pollen in each anther lobe of a stamen unites into a mass, forming a pollinium. Pollinia are attached to a clamp or clip like sticky structure called **corpusculum**. The filamentous or thread like part arising from each pollinium is called **retinaculum**. The

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